



Maturity and size estimation with yield mapping for hydroponic strawberries using machine vision

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ABSTRACT

Strawberry production in hydroponic systems requires precise and spatially explicit information on fruit load, maturity, and size to support harvest planning and quality control. This study presents an integrated computer vision and GNSS RTK framework that detects individual strawberries, estimates their maturity and physical dimensions, assigns fruits to operational size–maturity categories, and generates high-resolution yield maps in an open-field hydroponic system. RGB-D images are processed with YOLOv8-seg and a BoT-SORT-based tracking module to obtain instance-level masks, from which a color-based ripeness index and 3D point clouds are derived. After DBSCAN-based filtering, fruit length and width are computed in metric space and combined with ripeness percentage to assign each strawberry to one of nine operational size–maturity categories using predefined threshold rules. The system was evaluated in a commercial hydroponic crop in Arcabuco, Boyacá, Colombia, achieving accurate segmentation, with median IoU values up to 0.83 for bounding boxes and 0.71 for masks, and mean absolute errors of 2.51 mm in length and 1.85 mm in width with respect to Vernier caliper measurements. Yield maps aggregated in 1 m² cells and by crop row revealed marked spatial variability in fruit density, maturity state, and size–maturity composition, allowing the identification of zones with a high concentration of fruits ready for harvest versus areas dominated by immature fruits. The proposed framework provides a practical and low-cost decision support tool for hydroponic strawberry management and may be adapted to other high-value horticultural crops after recalibration under different crop architectures, cultivars, and environmental conditions.

1. Introduction

Accurate estimation of yield and maturity in high-value horticultural crops such as strawberries represents a fundamental technical and logistical challenge to improve harvest efficiency, reduce losses, and ensure product quality standards. In hydroponic systems, where plant density is higher and growing conditions are more controlled, data-driven planning becomes even more critical, especially in environments with limited labor or narrow commercial maturity windows [1]. However, continuous monitoring of fruit status remains predominantly manual, subjective, and costly.

In response to this need, the scientific community has made considerable progress in developing computer vision and deep learning systems to detect, classify, and characterize fruits across different crops. In a pioneering work, Swain et al. [2] developed a system to estimate yield in wild blueberries based on visual counts from images, achieving a correlation of $R^2 = 0.97$. More recently, Safre et al. [3] applied YOLOv8 and

YOLOv11 models for automatic tart cherry counting on conveyor belts, achieving a mAP@50 of 92% with YOLOv11x and a linear correlation with actual yield of $R^2 = 0.62$.

Beyond detection, various strategies have been developed to estimate maturity and size—two key attributes for commercial grading. Tamilarasi and Muthulakshmi [4] proposed a system to predict intermediate maturity in brinjal using k-means segmentation and contour analysis, reaching 96% accuracy and an F1-score of 91%. In fruits such as baobab, other studies have shown that morphological characteristics can be effectively modeled using computer vision and supervised methods, achieving volume predictions with $R^2 = 0.998$ [5].

Physical dimension estimation has also been the focus of several studies. Chu et al. [6] used a laser-camera scanner for three-dimensional localization in apples, achieving mean localization errors between 6.9 and 11.2 mm, significantly lower than those obtained with commercial RGB-D sensors like RealSense, which averaged 21.5 mm.

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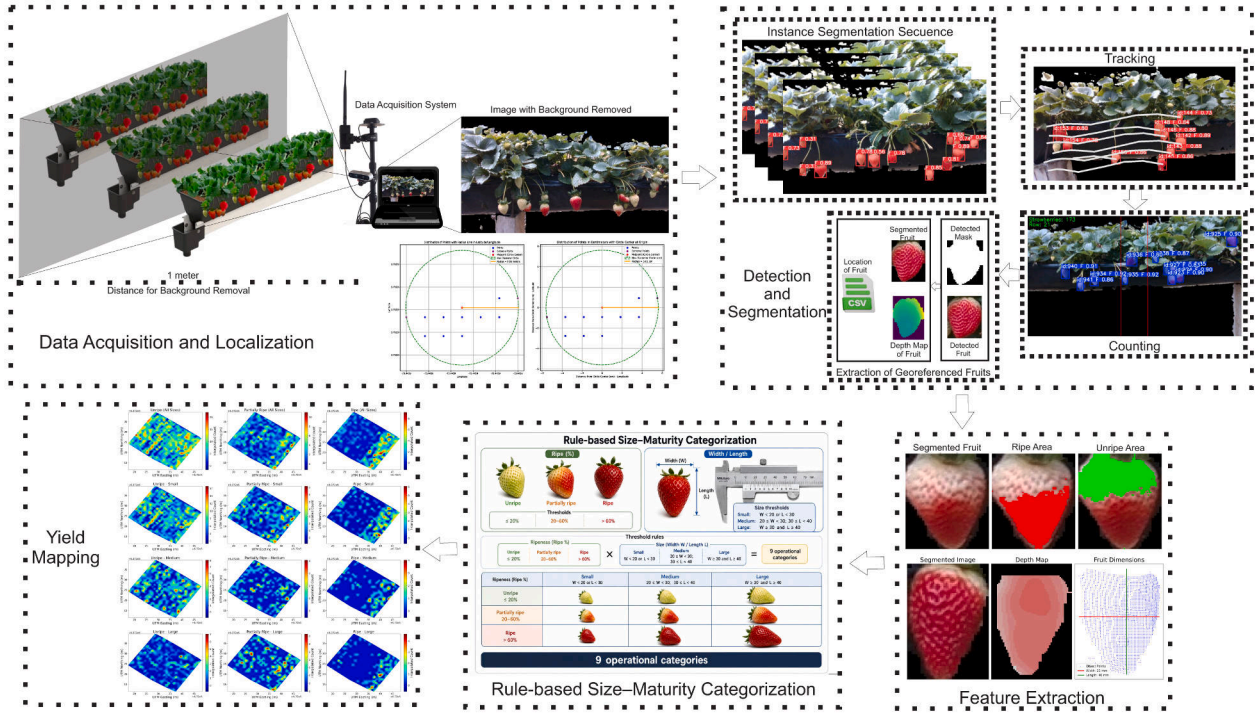


Fig. 1. Proposed framework.

Yield mapping has become a key tool in precision agriculture. Apolo-Apolo et al. [7] used UAVs to map citrus distribution, obtaining a prediction error of 6.59%, while [8] introduced a neural network that uses 16 visual fruit descriptors from olive trees to estimate yield with only a 2.64% overestimation compared to actual weight. Safre et al. [3] also integrated YOLO and BoT-SORT for tart cherry counting, validating the results with real weights and achieving an average error of 10 kg per tree.

Despite these advances, the simultaneous integration of segmentation, maturity/size estimation, and georeferenced fruit mapping remains limited, especially in low-structure crops like strawberries, where fruits occur in various maturity stages, with high occlusion and significant variability in color and size [9–11]. While some studies have applied the Segment Anything Model (SAM), Transformers, or multi-view strategies to improve segmentation and tracking [12,13], their application under real field conditions and their linkage to GNSS RTK coordinates remains incipient.

In the specific case of hydroponic crops, the literature is even more scarce. Sharmin et al. [1] reviewed yield prediction models for hydroponic lettuce, highlighting that most approaches lack spatial data integration, and that the production-control capabilities offered by hydroponics have not been fully exploited by computer vision systems. Similarly, studies on grapes [14], tomatoes [15], and bananas [16] underscore the need to model intra-plot variability in attributes such as maturity and weight to improve harvest planning, but these approaches have not been fully extended to structured hydroponic strawberry systems.

However, hydroponic strawberry crops should not be regarded merely as a planting method. From a computer vision and precision agriculture perspective, these systems introduce particular operating conditions: high fruit and plant density, fruit overlap, partial occlusion by leaves, support structures, repetitive backgrounds from adjacent rows, and illumination variability under open-field conditions. These characteristics increase scene complexity and complicate not only individual fruit detection, but also reliable maturity estimation, dimension measurement, and spatial localization of each fruit.

This work builds upon a previously developed RGB-D and GNSS RTK framework for hydroponic strawberry detection, segmentation, tracking, counting, and fruit-density yield mapping [17]. In the present study, that perception module is used as a validated input stage, and the main contribution is to extend it toward fruit-level characterization through maturity estimation, depth-based size measurement, operational size-maturity categorization, and enriched yield maps.

In this context, this study proposes a field-validated yield monitoring and mapping methodology for hydroponic strawberry crops. The system includes:

- Detection and segmentation of strawberries using deep learning models with RGB-D sensors.
- Estimation of fruit maturity through colorimetric analysis and fruit size through depth-based measurement.
- Georeferencing of each detected fruit using GNSS RTK coordinates.
- Rule-based assignment of fruits into operational size-maturity categories.
- Generation of yield maps in 1 m² grid cells with aggregated indicators per zone, including fruit count, maturity, size, and size-maturity composition.

This proposal addresses a key gap in the scientific literature by offering a practical, automated, and adaptable tool for spatially explicit fruit-level monitoring in hydroponic strawberry crops.

2. Materials and methods

Fig. 1 presents the framework devoted to strawberry feature extraction and yield mapping. Starting from RGB-D images georeferenced with GNSS RTK, the workflow is organized into five tightly coupled components: (i) data acquisition and localization, where RGB-D crop images are captured and associated with precise spatial coordinates; (ii) detection and segmentation, responsible for identifying and isolating each strawberry as an independent instance in the images; (iii) feature extraction, in which quantitative attributes are derived for each fruit, such as ripeness percentage and geometric dimensions; (iv) rule-based size-maturity categorization, which integrates ripeness and size information

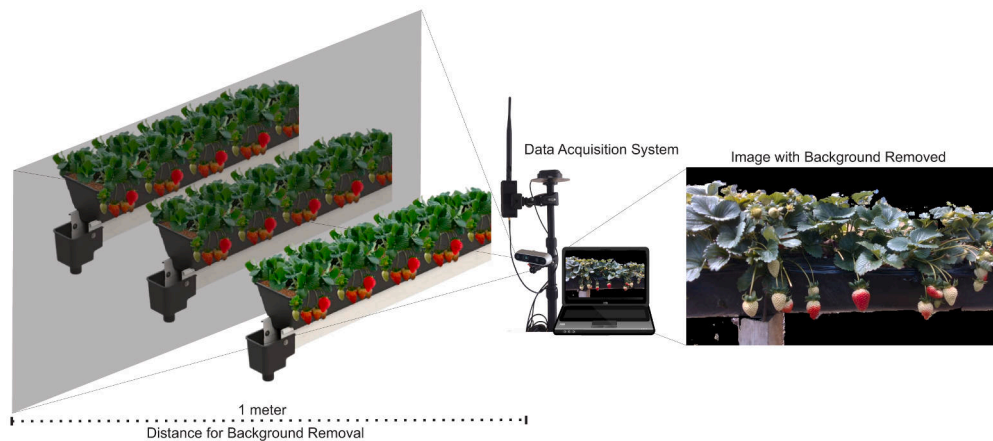


Fig. 2. Background removal.

to assign each fruit to a predefined operational category; and (v) yield mapping, where individual measurements are aggregated into discrete spatial units to build thematic maps of fruit count, maturity, size, and category composition. In the following subsections, each of these components is described in detail, following the order shown in the conceptual framework of the figure.

All modules described in this framework were implemented as an integrated processing workflow and applied to the field-collected dataset. In the current implementation, the recognition, tracking, feature extraction, size-maturity categorization, and map-generation stages are executed as a processing pipeline on the georeferenced RGB-D images. Therefore, the algorithms are not presented only as conceptual components, but as applied modules used to produce the measurements, categories, and maps analyzed in this study.

2.1. Data acquisition and localization

The proposed system was implemented in an open-field hydroponic strawberry crop located in Arcabuco, Boyacá, Colombia (Lat: 5°45'46.3" N, Lon: 73°26'46.9" W; altitude: 2739 m a.s.l.), using an Intel RealSense D435i RGB-D camera and an ArduSimple simpleRTK2B GNSS RTK system operating in a Base-Rover configuration. The camera and rover receiver were mounted on a telescopic monopod with a lateral view towards the crop rows, at an approximate height of 1.2 m above the ground, a lateral offset of about 0.6 m with respect to the row, and a tilt angle close to 25°. Data acquisition was carried out in the morning (between 8:00 and 11:00 h), under partially cloudy conditions, in order to reduce abrupt changes in illumination. Under these conditions, the plants exhibited different fruiting stages, with fruits of various sizes and ripeness levels. A total of 1626 RGB-D images were captured at a resolution of 1280 × 720 pixels and a frame rate of 7 fps, which allowed dense coverage of the crop and adequate spatial continuity for subsequent analysis stages.

Camera control and data flow management were performed in Python using the `pyrealsense2` library and Intel RealSense SDK 2.0 tools. The detailed preprocessing of depth images (application of decimation filters, disparity-depth transformations, spatial and temporal filters, and hole filling) has been described in a previous work by the authors on strawberry detection and counting using georeferenced RGB-D images in the same hydroponic system [17]. In the present article, this preprocessing is assumed as an input stage, and one additional step specifically related to feature extraction is highlighted.

Fig. 2 illustrates the background removal process, where the already filtered depth information is used to discard pixels that fall outside the expected range of the target crop row. Based on the depth map, a distance interval associated with the furrow where the plants and fruits of the analyzed row are located is defined. Pixels with depth values below

or above this interval are marked as background and masked out, thus removing distant rows, support structures and other elements that are not part of the fruit analysis. Simple morphological filtering operations are applied to clean small disconnected regions, so that the resulting image mainly preserves the foliage and fruits of the row of interest. This procedure reduces scene complexity and improves the robustness of the subsequent detection, segmentation and feature extraction stages, in a manner consistent with other vision-based fruit yield mapping systems that integrate RGB-D sensors [18,19] and GNSS RTK [20,21].

In parallel with RGB-D image acquisition, localization data from the GNSS RTK system on the simpleRTK2B rover were continuously recorded. The base receiver was placed at a point with known coordinates and transmitted real-time differential corrections to the rover through a radio link in the 902–927 MHz band. This Base-Rover configuration compensates for atmospheric and geometric signal errors, achieving centimetre-level positioning accuracy. The acquisition system was programmed with independent execution threads for the GNSS and the camera, synchronized by shared timestamps so that each RGB-D image was associated with a corresponding GNSS RTK position.

To quantify the effective accuracy of the GNSS RTK system under the actual operating conditions, a static experiment was carried out by keeping the rover at a fixed point for an extended period. The resulting localization coordinates were represented in a planar coordinate system, generating a dispersion pattern around the true position. Fig. 3 shows the dispersion of GNSS data at a fixed location, with the resulting point cloud and reveals a maximum radius of approximately 5.61 cm, which characterizes the variability of the system in Base-Rover mode. This dispersion is relevant because it directly propagates to the spatial accuracy of the fruit-level yield maps.

Although this centimeter-level accuracy characterizes the localization of the acquisition platform and propagates to the yield maps, it should not be interpreted as the ability to spatially resolve adjacent fruits separated by only a few centimeters within the same plant or fruit cluster. In this system, individual fruit separation is primarily performed in image space through instance segmentation and temporal tracking, whereas GNSS RTK provides the geospatial reference for each image and its associated detections. Therefore, RTK accuracy is adequate for the target spatial scale of this study, namely crop-row analysis and aggregation in 1 m² grid cells, but it is not intended to replace visual discrimination of closely spaced fruits within the canopy.

Finally, the GNSS coordinates were transformed from latitude-longitude to a projected coordinate system in metres, specifically UTM (Universal Transverse Mercator), as commonly employed in precision agriculture applications [22,23]. This transformation allows each image and each detected fruit to be located within a Cartesian reference frame suitable for spatial analysis and yield map generation. The acquisition and localization scheme described in this section forms the basis

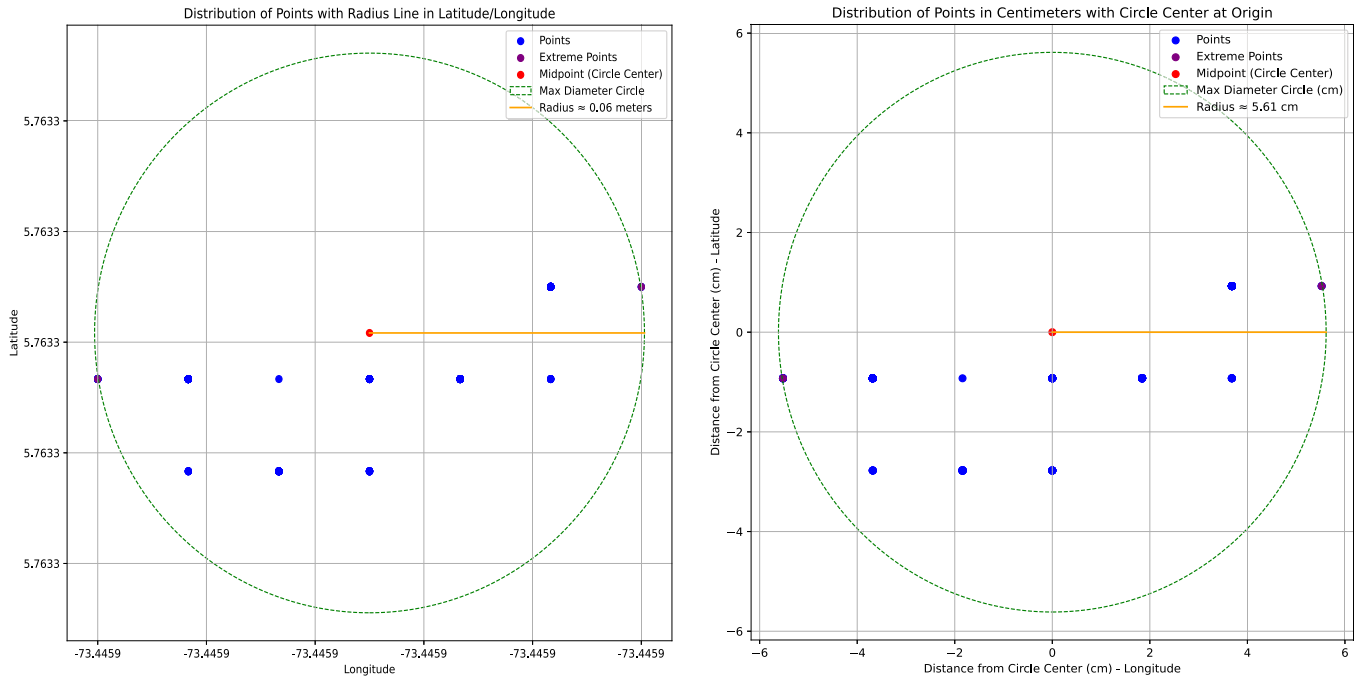


Fig. 3. Dispersion of GNSS data at a fixed location.

on which the subsequent stages of detection and segmentation, feature extraction and yield mapping presented in this work are built.

2.2. Detection and segmentation

The detection and segmentation stage receives as input the RGB-D images that have already undergone the preprocessing and background removal described in the previous section. On these images, an instance segmentation pipeline is implemented, allowing each strawberry to be individually identified at the pixel level and associated with its corresponding depth information and geographic location.

For strawberry detection, four variants of the YOLOv8 model with instance segmentation capabilities were used: YOLOv8n-seg, YOLOv8s-seg, YOLOv8m-seg, and YOLOv8l-seg. The construction of the dataset, the annotation protocol, the partition into training, validation, and test subsets, and the training details of these models were presented in detail in [17]. In the present manuscript, YOLOv8-seg is not introduced as the main algorithmic novelty, but as a previously developed perception module that provides fruit masks, bounding boxes, tracking identifiers, and georeferenced detections. The methodological contribution of this study is to extend this perception stage toward fruit-level feature extraction, maturity and size estimation, operational size-maturity categorization, and enriched yield mapping.

Dataset annotations were performed at the instance level, delineating each visible strawberry in the image with polygons that follow the fruit contour. From these polygons, the corresponding bounding boxes are also derived. This dual representation (polygon masks and bounding boxes) enables the evaluation of both object detection performance and instance segmentation quality.

During inference, each preprocessed RGB image is fed into one of the YOLOv8-seg models. For each detected strawberry, the network outputs the class label (strawberry), a confidence score, a bounding box, and a binary mask that delineates the fruit at pixel level. The resulting set of detections per image constitutes the input for the temporal tracking module. Tracking of strawberries along the image sequence is performed using a tracking-by-detection strategy based on the BoT-SORT algorithm, which associates detections across consecutive frames by combining motion and appearance information. Each strawberry is

assigned a track identifier (track ID) when it is detected for the first time, and this identifier is maintained while the fruit remains visible. If a track stops receiving associations for a predefined number of frames, it is considered terminated. This mechanism prevents double counting when the same strawberry appears in multiple consecutive frames with small viewpoint changes or partial occlusions.

The Fig. 4 summarizes the overall operation of this stage. In the initial part of the figure, a set of RGB images is shown after depth filtering and background removal, so that only the target crop row remains visible. The figure also shows the result of instance segmentation, where strawberries are detected with their corresponding masks and bounding boxes. Subsequently, the tracking process is depicted, where each strawberry is followed along the sequence by its track ID, ensuring that the system identifies it as the same object in different frames and enabling a unique count for each fruit. Finally, the figure illustrates the individual extraction of strawberries: for each detection, the RGB image is cropped using the mask, the aligned depth map is cropped in the same region, and the associated information is organized in a structured file system.

More specifically, for each detected strawberry, three main types of information are stored:

- The segmented color image of the fruit, obtained by cropping the fruit region in the RGB image using its instance mask.
- The depth map corresponding to the fruit, cropped from the original depth image aligned with the RGB image, using the same instance mask.
- The detection metadata, including the mask (as a binary image), the track identifier (track ID), and the georeferenced location of the fruit, obtained from the GNSS RTK system.

The localization information of each fruit is stored in a .csv file, where, among other fields, the image identifier, the track ID, the projected coordinates (UTM), and additional labels required for subsequent stages are recorded. This procedure produces a dataset of georeferenced strawberry instances, in which each fruit is represented by its RGB appearance, its local 3D structure (through depth), and its spatial position within the crop. These data form the basis for the feature extraction stage (Section 2.3) and the yield mapping stage (Section 2.5).

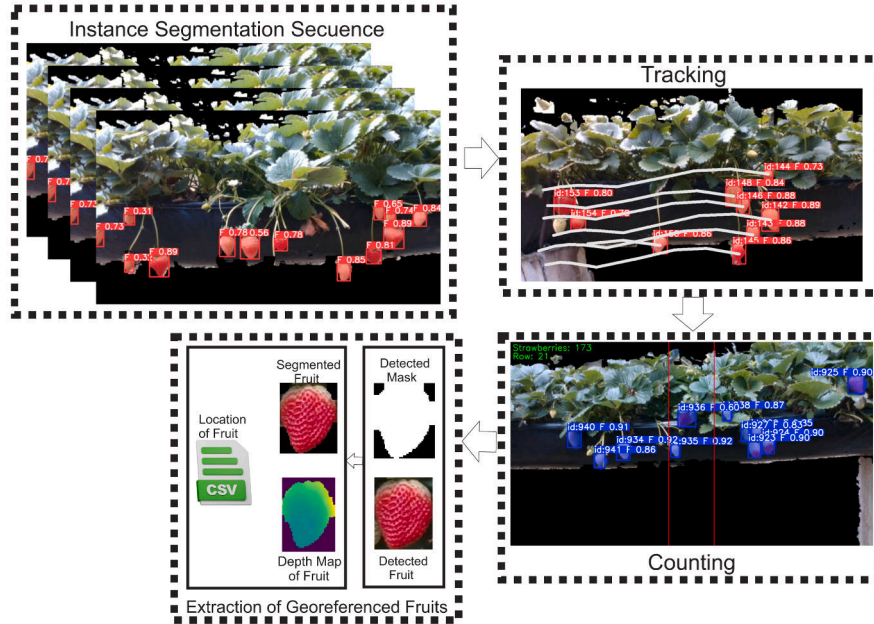


Fig. 4. Segmentation and extraction of georeferenced fruits.

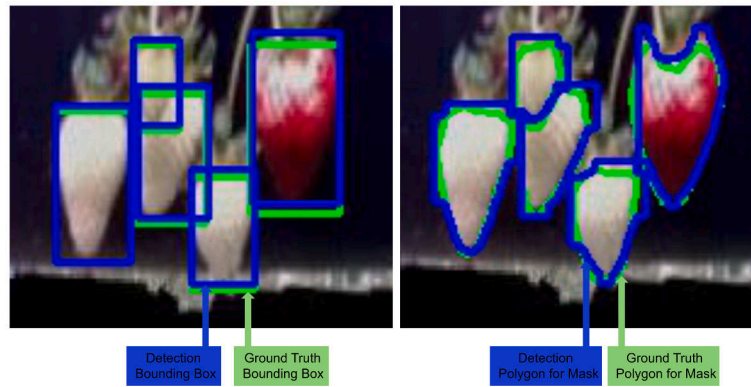


Fig. 5. Qualitative comparison between predicted detections and ground-truth annotations for bounding boxes and instance masks.

2.2.1. Evaluation metrics

To quantitatively evaluate detection and segmentation performance, geometric metrics were computed on the test subset for both bounding boxes and instance masks. Specifically, F1-Score, interpreted as the Dice coefficient between the predicted and ground-truth regions, and Intersection over Union (IoU), which measures the overlap ratio between both regions, were used. These metrics were computed for each prediction-ground-truth pair and summarized using the median, mean, and standard deviation to characterize both typical performance and model variability under different scene configurations, illumination conditions, and occlusion levels.

In addition to detection and segmentation quality, the computational performance of each YOLOv8-seg variant was evaluated using total inference time, average inference time per image, frames per second (FPS), and peak GPU memory usage. These metrics allow the geometric accuracy and computational efficiency of YOLOv8n-seg, YOLOv8s-seg, YOLOv8m-seg, and YOLOv8l-seg to be compared. Fig. 5 qualitatively illustrates the comparison between predictions and ground-truth annotations for bounding boxes and masks.

Overall, the detection and segmentation stage establishes the link between the preprocessed RGB-D sequences and the subsequent modules for feature extraction, rule-based size-maturity categorization, and

yield mapping. This stage ensures that each strawberry is localized, segmented, tracked over time, and associated with its depth information and georeferenced location.

2.3. Feature extraction

For each strawberry detected and segmented in the previous stage, the system extracts two main groups of features:

- A ripeness indicator, associated with the predominance of red tones on the fruit surface.
- The physical dimensions of each strawberry (width and length, in millimeters), computed from the depth map and the intrinsic parameters of the depth camera.

These features are individually computed for each fruit and stored together with its georeferenced location, forming the basis for the subsequent operational size-maturity categorization and for the construction of yield maps.

Methodologically, feature extraction relies on two key inputs produced by the detection and segmentation stage:

- The segmented RGB image of the strawberry, where the background has been removed and only the fruit remains.

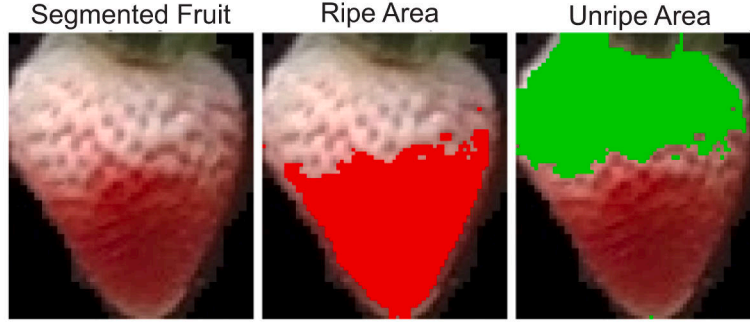


Fig. 6. Ripeness determination.

- The aligned depth submatrix corresponding to that fruit, containing the distance values for the fruit pixels.

From these inputs, colour masks in HSV space are built to estimate the percentage of ripe and unripe area, and a 3D point cloud of the fruit is generated from the depth values and the camera intrinsics. A density-based clustering algorithm (DBSCAN) is then applied to this point cloud to filter out distant or noisy points before computing the fruit dimensions.

The following subsections describe in detail the procedure for ripeness determination (Section 2.3.1) and for the estimation of the fruit size (Section 2.3.2), including the mathematical formulation and the specific role of DBSCAN in obtaining dimension estimates.

2.3.1. Ripeness determination

Ripeness is estimated in HSV colour space, taking advantage of its ability to separate the hue component from luminance and saturation. The input to this stage is the segmented fruit image (without background) obtained after detection and segmentation.

Let I_{RGB} denote the RGB image of the segmented strawberry and I_{HSV} its representation in HSV space. A binary “fruit area” mask $mask_{fruit}$ is obtained by applying an initial HSV threshold to roughly isolate the fruit, followed by a logical inversion and a morphological erosion to remove small noisy regions. Conceptually:

$$mask_{fruit}(p) = \begin{cases} 1, & \text{if pixel } p \text{ is considered part of the fruit,} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Within $mask_{fruit}$, two types of regions are identified: *ripe* (red) and *unripe* (green/yellow).

- Ripe regions (red). Two hue intervals are used to cover red tones in HSV:

$$H \in [0, 10], \quad S \geq 100, \quad V \geq 100, \quad (2)$$

$$H \in [170, 180], \quad S \geq 100, \quad V \geq 100. \quad (3)$$

From these intervals, partial masks $mask_{mature_1}$ and $mask_{mature_2}$ are obtained, and the global ripe mask is defined as:

$$mask_{mature} = (mask_{mature_1} \vee mask_{mature_2}) \wedge mask_{fruit}. \quad (4)$$

- Unripe regions (green/yellow). Unripe areas are identified with a hue interval for green/yellow tones:

$$H \in [25, 85], \quad S \geq 40, \quad V \geq 40, \quad (5)$$

leading to a provisional mask $mask_{immature_HSV}$, which is then restricted to the fruit area:

$$mask_{immature} = mask_{immature_HSV} \wedge mask_{fruit}. \quad (6)$$

The HSV thresholds used in this study are fixed thresholds defined for the evaluated acquisition conditions. Although HSV partially separates chromatic information from intensity, the results may still be affected

by illumination changes, shadows, specular reflections, and exposure variations. To reduce these effects, the analysis is applied only to the previously segmented fruit region rather than to the complete image, thereby limiting the influence of the background and crop structures. In addition, strawberry achenes may introduce yellow or light-colored pixels within the fruit surface; therefore, the ripeness index is interpreted as a relative measure of visible red-surface coverage, rather than as a direct physiological measurement of internal ripeness. In future implementations, this stage could be improved through adaptive color calibration, radiometric normalization, or learning-based ripeness estimation models.

Pixels in the fruit area that do not fall into either ripe or unripe ranges are first marked as unclassified,

$$mask_{unclassified} = mask_{fruit} \wedge \neg(mask_{mature} \vee mask_{immature}), \quad (7)$$

and then reassigned to the unripe mask:

$$mask_{immature} \leftarrow mask_{immature} \vee mask_{unclassified}. \quad (8)$$

Ripeness percentage. Define

$$M = \{p \mid mask_{mature}(p) = 1\}, \quad (9)$$

$$U = \{p \mid mask_{immature}(p) = 1\}, \quad (10)$$

with

$$N_{mature} = |M|, \quad N_{immature} = |U|. \quad (11)$$

The percentage of ripe area is:

$$Ripe = \begin{cases} \frac{N_{mature}}{N_{mature} + N_{immature}} \times 100, & \text{if } N_{mature} + N_{immature} > 0, \\ \text{NaN}, & \text{otherwise,} \end{cases} \quad (12)$$

and the unripe percentage is:

$$Unripe = 100 - Ripe. \quad (13)$$

Thus, each strawberry is characterized by the pair (Ripe, Unripe), which quantifies its surface ripeness as the proportion of red area over the effective fruit area.

This process is illustrated in Fig. 6, which shows the segmented RGB fruit without background, the ripe area highlighted in red, and the unripe area highlighted in green.

2.3.2. Determination of fruit size (Length and width)

The physical dimensions of each strawberry (length and width, in millimeters) are estimated from the depth map aligned with the segmented fruit image and from the intrinsic parameters of the Intel RealSense D435i camera. The goal is to build a 3D point cloud of each fruit, filter spurious points using DBSCAN, and compute its main geometric extents.

• Projection to 3D coordinates.

For each segmented strawberry, the corresponding depth submatrix, aligned with the RGB image, is masked using the combined fruit mask obtained from the ripeness stage, $\text{combined_mask} = \text{mask}_{\text{mature}} \vee \text{mask}_{\text{immature}}$, so that only depth values belonging to the fruit are retained. All pixels with valid depth are projected to metric coordinates (X, Y, Z) in the camera frame using a pinhole model and the calibrated intrinsic parameters of the Intel RealSense D435i:

$$f_x = 1210, \quad f_y = 1210, \quad c_x = 640, \quad c_y = 360, \quad (14)$$

$$X = \frac{(j - c_x)z}{f_x}, \quad Y = -\frac{(i - c_y)z}{f_y}, \quad Z = z, \quad (15)$$

where z is the depth (in millimeters) at pixel (i, j) . This process yields a 3D point cloud Q that represents the fruit geometry from the camera viewpoint.

In Fig. 7, subfigure (a) shows the segmented RGB strawberry, while subfigure (b) shows the associated depth map from which Q is derived.

• Outlier filtering with DBSCAN and its effect on dimensions.

Despite 2D segmentation, Q may contain points that do not belong to the actual fruit body (e.g., mis-segmented background, leaves, support structures, reflections). Using the full cloud directly could artificially inflate the extensions along X and Y , leading to unrealistic size estimates.

To avoid this, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm is applied to the points in Q . DBSCAN groups points according to local density and labels as noise those that do not belong to dense regions. In the implementation, the following parameters are used:

$$\epsilon = 5 \text{ mm}, \quad m = 10, \quad (16)$$

where ϵ is the neighbourhood radius (eps) and m is the minimum number of points (min_samples). The largest non-noise cluster is selected as the main fruit cluster, and all remaining points are treated as outliers and discarded before dimension estimation.

This behaviour is illustrated in Fig. 7:

- In subfigure (c), *Fruit Dimensions Raw*, a 2D view of the original point cloud is shown. A compact region corresponds to the main fruit body, while a set of distant points appears highlighted in red.
- In subfigure (d), *Fruit Dimensions DBSCAN*, the same 2D view is shown after applying DBSCAN. The distant red points are removed as noise and only the compact cluster remains.
- Subfigures (e), *Fruit Dimensions Raw (3-D)*, and (f), *Fruit Dimensions DBSCAN (3-D)*, present 3D views of the raw and filtered point clouds, respectively, clearly showing the dispersion of distant points and the compact volume of the fruit after DBSCAN filtering.

This figure highlights the role of DBSCAN in the size feature extraction stage: by systematically removing distant depth points that do not belong to the fruit body, it prevents length and width overestimation and yields geometrically consistent measurements.

• Computation of width and length from the main cluster.

Once the main cluster has been obtained, the fruit dimensions are computed along the X (width) and Y (length) axes. From all points in main cluster, the minimum and maximum coordinates are obtained, and the raw geometric extensions along each axis are:

$$\Delta X = X_{\max} - X_{\min} \quad \Delta Y = Y_{\max} - Y_{\min} \quad (17)$$

To compensate for systematic biases associated with viewing geometry, depth discretization, partial visibility of the fruit surface, and spatial sampling of the point cloud, the geometric extensions were adjusted using empirical scale factors. These factors were obtained during a calibration stage using manual reference measurements with a Vernier caliper and were kept fixed for processing the

evaluated dataset. In the implementation, the following scale factors were used:

$$s_w = 1.43 \quad (\text{width scale factor}), \quad s_l = 1.67 \quad (\text{length scale factor}). \quad (18)$$

The calibrated continuous estimates are computed as:

$$\widehat{W} = \Delta X \cdot s_w, \quad \widehat{L} = \Delta Y \cdot s_l. \quad (19)$$

For tabular reporting and subsequent operational categorization, the dimensions were rounded to the nearest millimeter:

$$\text{Fruit_Width} = \text{round}(\widehat{W}), \quad \text{Fruit_Length} = \text{round}(\widehat{L}). \quad (20)$$

This rounding avoids reporting sub-millimetric precision that is not justified under field conditions with an RGB-D sensor, without changing the practical interpretation of the size classes defined in this study.

In the example illustrated in Fig. 7, before filtering with DBSCAN, the raw extensions produce $W = 39 \text{ mm}$ and $L = 39 \text{ mm}$, whereas after filtering and recomputing the extensions from main cluster only, the dimensions adjust to $W = 29 \text{ mm}$ and $L = 38 \text{ mm}$.

Although DBSCAN filtering reduces the influence of spurious points and mis-segmented background regions, size estimation may still be affected by partial occlusions, incomplete masks, depth quantization, sensor noise, and the limited visibility of the fruit from a single lateral viewpoint. For this reason, the estimated dimensions should be interpreted as practical measurements obtained from the camera perspective, rather than as complete geometric reconstructions of the fruit.

• Validation of dimension measurements.

To validate the dimensions obtained by the vision algorithm, a comparison experiment was conducted against manual measurements using a Vernier caliper. A set of strawberries was measured manually (width and length) and processed with the algorithm described above.

For fruit i , let:

- L_i^{ref} and W_i^{ref} : reference length and width measured with the caliper,
- L_i^{cv} and W_i^{cv} : length and width estimated by the vision system.

The differences between both measurements are quantified using standard error metrics. For length:

- **Mean Absolute Error (MAE) in length:**

$$\text{MAE}_L = \frac{1}{N} \sum_{i=1}^N |L_i^{\text{cv}} - L_i^{\text{ref}}|. \quad (21)$$

- **Mean Relative Error (MRE) in length:**

$$\text{MRE}_L = \frac{1}{N} \sum_{i=1}^N \frac{|L_i^{\text{cv}} - L_i^{\text{ref}}|}{L_i^{\text{ref}}} \times 100. \quad (22)$$

- **Root Mean Squared Error (RMSE) in length:**

$$\text{RMSE}_L = \sqrt{\frac{1}{N} \sum_{i=1}^N (L_i^{\text{cv}} - L_i^{\text{ref}})^2}. \quad (23)$$

Analogous metrics MAE_W , MRE_W and RMSE_W are defined for width. The specific values of these metrics are reported in the Results section, showing that the errors remain within acceptable ranges for field applications in precision agriculture.

This validation procedure is summarised in Fig. 8:

- on the left, strawberries being measured with the Vernier caliper are shown;
- in the centre, the same fruits are captured by the RGB-D camera;
- on the right, the length and width values estimated by the vision algorithm are displayed, highlighting the general agreement between manual and vision-based measurements.

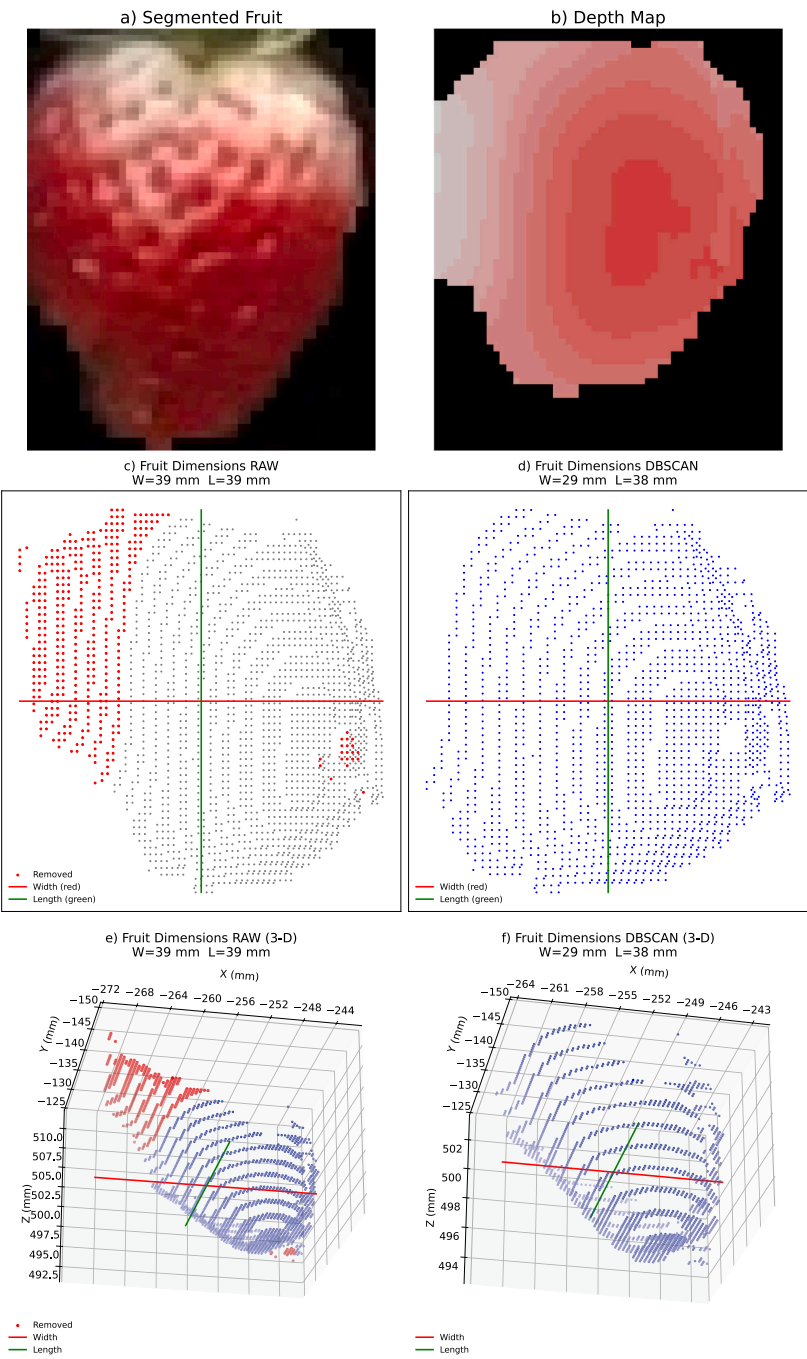


Fig. 7. Depth map and fruit dimensions with DBSCAN.

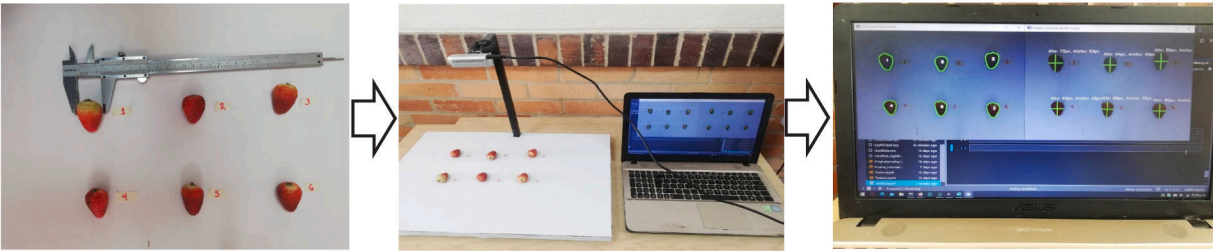


Fig. 8. Validation of dimension measurements: Vernier caliper and computer vision algorithm.

Table 1

Distribution of strawberries by operational size–maturity category in the original dataset.

Category ID	Size–maturity category	Number of fruits
0	Partially Ripe–Large	178
1	Partially Ripe–Medium	196
2	Partially Ripe–Small	635
3	Ripe–Large	150
4	Ripe–Medium	147
5	Ripe–Small	379
6	Unripe–Large	352
7	Unripe–Medium	879
8	Unripe–Small	2670

• Feature set.

Finally, for each strawberry, the extracted feature set consists of:

- ripeness and unripeness percentages, Ripe and Unripe,
- physical dimensions, Fruit_Width and Fruit_Length (in mm),
- georeferenced location information (Coordinates associated with the fruit).

These data are integrated into a tabular structure in CSV format and form the basis for the subsequent stages of rule-based operational size–maturity categorization (Section 2.4) and yield mapping (Section 2.5).

2.4. Rule-based assignment of strawberry size–maturity categories

Once the features of each fruit, specifically ripeness percentage and physical dimensions, have been computed, each strawberry is assigned to one of nine operational categories that combine maturity and size information. This stage uses a transparent rule-based categorization procedure, in which the extracted fruit descriptors are converted into discrete agronomic classes to support spatial interpretation, yield mapping, and harvest planning.

The categorization is based on two main axes:

- **Ripeness**, expressed as the percentage of ripe fruit area obtained from the masks of ripe and unripe regions (Section 2.3.1). From this continuous value, three levels are defined:
 - *Unripe*: ripeness $\leq 20\%$,
 - *Partially Ripe*: $20\% < \text{ripeness} \leq 60\%$,
 - *Ripe*: ripeness $> 60\%$.
- **Size**, derived from the physical dimensions Fruit_Width and Fruit_Length (in mm), as described in Section 2.3.2. Three size ranges are defined:
 - *Small*: width < 20 mm or length < 30 mm,
 - *Medium*: $20 \text{ mm} \leq \text{width} < 30 \text{ mm}$ and $30 \text{ mm} \leq \text{length} < 40 \text{ mm}$,
 - *Large*: width $\geq 30 \text{ mm}$ and length $\geq 40 \text{ mm}$.

Combining the three ripeness levels with the three size ranges yields nine size–maturity categories. Each fruit receives a category identifier between 0 and 8, which enables aggregation by spatial cell and crop row during yield mapping. These categories are used as operational indicators to summarize spatial crop variability, distinguishing zones dominated by developing immature fruits from zones with larger ripe fruits of greater interest for selective harvesting.

The thresholds used in this study should be interpreted as operational criteria defined for the evaluated monitoring system, rather than as universal commercial standards. They were selected to produce agronomically interpretable groups compatible with the measurement scale of the RGB-D system. When the system is transferred to other production contexts, the thresholds can be recalibrated or aligned with cultivar-specific traits, local commercial grading criteria, grower requirements, or postharvest grading standards. The distribution of fruits per category in the original dataset is shown in Table 1.

2.5. Yield mapping

The yield mapping stage starts from the final set of detected strawberries

$$F = \{f_k\}_{k=1}^N, \quad (24)$$

where each fruit f_k is described by:

- projected coordinates (x_k, y_k) in UTM (meters),
- ripeness percentage r_k (% of visible ripe area),
- physical dimensions (w_k, l_k) (width and length, in mm),
- rule-based operational size–maturity category $c_k \in \{0, \dots, 8\}$, as defined in Section 2.4.

The goal is to transform these point-level data into spatial indicators that allow crop variability to be analyzed in terms of fruit density, visible ripeness, physical dimensions, and size–maturity category composition.

2.5.1. Aggregation in 1 m² cells

Over the plot, expressed in UTM coordinates, a regular grid of square cells C_{ij} of 1 m $\times$ 1 m is defined. Each strawberry is assigned to the cell that contains its point (x_k, y_k) , so that for each cell the set of fruits F_{ij} and its size are obtained:

$$N_{ij} = |F_{ij}|, \quad (25)$$

which directly corresponds to the fruit density (fruits/m²) at that location in the crop.

From the fruits contained in each cell, the following indicators are computed:

- **Fruit density**: the value N_{ij} is used to generate density maps, where each 1 m² cell represents how many strawberries are found in that area. This makes it possible to identify zones with higher and lower fruit load.
- **Average ripeness per cell**: using the ripe area percentage r_k of each fruit $f_k \in F_{ij}$, the average ripeness of the cell is obtained as the mean of those values:

$$\bar{r}_{ij} = \begin{cases} \frac{1}{N_{ij}} \sum_{f_k \in F_{ij}} r_k, & N_{ij} > 0, \\ \text{NaN}, & N_{ij} = 0, \end{cases} \quad (26)$$

which reflects the average ripeness status in each area of the crop.

- **Category distribution**: for each operational size–maturity category $h \in \{0, \dots, 8\}$, the number of fruits of that category in the cell is counted:

$$N_{ij}^{(h)} = |\{f_k \in F_{ij} \mid c_k = h\}|, \quad (27)$$

so that the vector $(N_{ij}^{(0)}, \dots, N_{ij}^{(8)})$ describes the composition of the cell in terms of size–maturity categories, for example, how many “Ripe–Large”, “Unripe–Small”, etc. fruits are concentrated in each square meter.

When a smoother representation of these spatial patterns is required, the values computed per cell, including total density, average ripeness, or counts by size–maturity category, are associated with the cell centroids and interpolated over a denser grid. The result is a set of continuous surfaces that allow gradients and transition zones in fruit load and crop composition to be visualized at the plot scale. These interpolated maps are used only as visual aids for interpreting spatial trends. Quantitative interpretation is based on the original counts and averages computed in the 1 m² cells, since interpolation may produce smoothed, non-integer, or intermediate values that do not correspond to direct fruit-count observations.

Table 2
Performance of YOLOv8-seg models on the test dataset for bounding boxes and masks.

Metric	Bounding Boxes				Masks			
	YOLOv8n-seg	YOLOv8s-seg	YOLOv8m-seg	YOLOv8l-seg	YOLOv8n-seg	YOLOv8s-seg	YOLOv8m-seg	YOLOv8l-seg
F1-Score (Median)	0.81	0.83	0.87	0.90	0.73	0.75	0.78	0.80
F1-Score (Mean)	0.78	0.80	0.83	0.85	0.70	0.73	0.75	0.77
F1-Score (Std Dev)	0.15	0.14	0.14	0.13	0.16	0.15	0.15	0.14
IoU (Median)	0.73	0.76	0.80	0.83	0.63	0.65	0.69	0.71
IoU (Mean)	0.70	0.73	0.77	0.79	0.61	0.64	0.66	0.68
IoU (Std Dev)	0.16	0.15	0.14	0.13	0.15	0.15	0.15	0.14
Total Inference Time (s)	25.64	24.94	39.46	64.53	34.65	35.89	44.16	65.15
Inference Speed (ms)	29.00	28.21	44.63	73.00	39.20	40.60	49.96	73.70
FPS	34.48	35.45	22.40	13.70	25.51	24.63	20.02	13.57
GPU Memory Used (MB)	114.48	94.35	106.72	123.45	139.67	81.92	92.42	110.33

2.5.2. Row-wise distribution analysis

In addition to the analysis based on regular cells, a specific study of category distribution along the crop rows is carried out. Each fruit f_k is associated with a row index $\text{Row}(f_k)$. For each row r , the following are defined:

- the set of fruits belonging to row r ,
- the total number of fruits N_r ,
- the counts per category $N_{r,h}$, which represent how many fruits of category h are found in row r .

Formally, the total number of fruits in row r and the counts per category are expressed as:

$$N_r = \left| \{f_k \in F \mid \text{Row}(f_k) = r\} \right|, \quad (28)$$

$$N_{r,h} = \left| \{f_k \in F \mid \text{Row}(f_k) = r, c_k = h\} \right|. \quad (29)$$

From these counts, row and category-wise proportions are computed:

$$p_{r,h} = \begin{cases} \frac{N_{r,h}}{N_r}, & N_r > 0, \\ 0, & N_r = 0, \end{cases} \quad (30)$$

which indicate what fraction of the fruits in row r belongs to each of the nine categories considered.

The matrix formed by $N_{r,h}$ (or by $p_{r,h}$) is represented using heatmap-type plots, where:

- each row in the plot corresponds to a crop row,
- each column represents a strawberry category,
- the color intensity reflects the count or proportion associated with each row–category combination.

This type of analysis makes it possible to identify rows with a higher presence of large ripe fruits, rows dominated by small unripe fruits, or more balanced patterns. Overall, the yield mapping stage formalizes the transition from individual fruit data $(x_k, y_k, r_k, w_k, l_k, c_k)$ to spatial yield indicators aggregated by 1 m² cells and by crop rows, which serve as the basis for the agronomic interpretation presented in the Results and Discussion section.

3. Results and discussion

3.1. Evaluation of performance metrics on the test dataset

The results in this section summarize the performance of the detection and segmentation module used as the perception stage of the system. Since the original training and validation of the YOLOv8-seg models were previously presented, these metrics are reported here to contextualize the quality of the masks, bounding boxes, and detections that feed the new stages of this manuscript: maturity estimation, size

measurement, rule-based size–maturity categorization, and yield mapping.

Table 2 summarizes the quantitative performance of the four YOLOv8-seg variants on the test dataset, for both *bounding boxes* and *segmentation masks*. The metrics reported follow the definitions introduced in Section 2.2.1 and are complemented by the boxplots in Fig. 9, which illustrate the distribution of F1-Score and IoU values across all test instances.

For bounding boxes, a clear trend is observed as model complexity increases from YOLOv8n-seg to YOLOv8l-seg. The median F1-Score increases from 0.81 to 0.90, while the median IoU rises from 0.73 to 0.83. Similarly, the mean values improve (F1-Score from 0.78 to 0.85, IoU from 0.70 to 0.79), indicating that larger models not only achieve better best-case predictions, but also increase the average overlap between predicted and ground-truth regions. The standard deviations for F1-Score and IoU remain within a relatively narrow range (approximately 0.13–0.16), suggesting that these gains are not obtained at the cost of higher instability in more complex scenes. The upper row of Fig. 9 shows that the interquartile ranges tend to shrink for YOLOv8m-seg and YOLOv8l-seg, with higher medians and fewer low outliers, consistent with more reliable localization of strawberry instances.

For segmentation masks, the same monotonic trend holds, although with slightly lower absolute values, as expected for a more demanding task than box-level detection. The median F1-Score for masks increases from 0.73 (YOLOv8n-seg) to 0.80 (YOLOv8l-seg), and the median IoU from 0.63 to 0.71; in parallel, the mean F1-Score and IoU improve from 0.70 to 0.77 and from 0.61 to 0.68, respectively. These results indicate that the segmentation head of YOLOv8-seg can capture fruit contours with good accuracy despite occlusions, overlapping fruits, and appearance variability. The lower row of Fig. 9 shows that, although the dispersion of mask-based metrics is somewhat higher than for bounding boxes, the distributions corresponding to YOLOv8m-seg and YOLOv8l-seg are clearly shifted toward higher overlap values, confirming their superior ability to delineate the effective fruit area used later for ripeness and size estimation.

In addition to detection and segmentation quality, Table 2 reports the computational performance of each model. As expected, there is a clear trade-off between accuracy and inference speed. For bounding boxes, YOLOv8n-seg and YOLOv8s-seg achieve the highest processing rates, with 34.48 and 35.45 FPS, and mean inference times close to 28–29 ms per image. In contrast, YOLOv8m-seg and YOLOv8l-seg reduce the processing rate to 22.40 and 13.70 FPS, with inference times of 44.63 and 73.00 ms per image. A similar behavior is observed for masks, where the additional cost of the segmentation head further decreases FPS (from 25.51 FPS for YOLOv8n-seg to 13.57 FPS for YOLOv8l-seg). GPU memory usage remains within moderate ranges for all configurations (on the order of tens to a few hundred MB), which facilitates deployment on standard GPU platforms.

From an application standpoint, these results define different operating points depending on the requirements of the target system. If

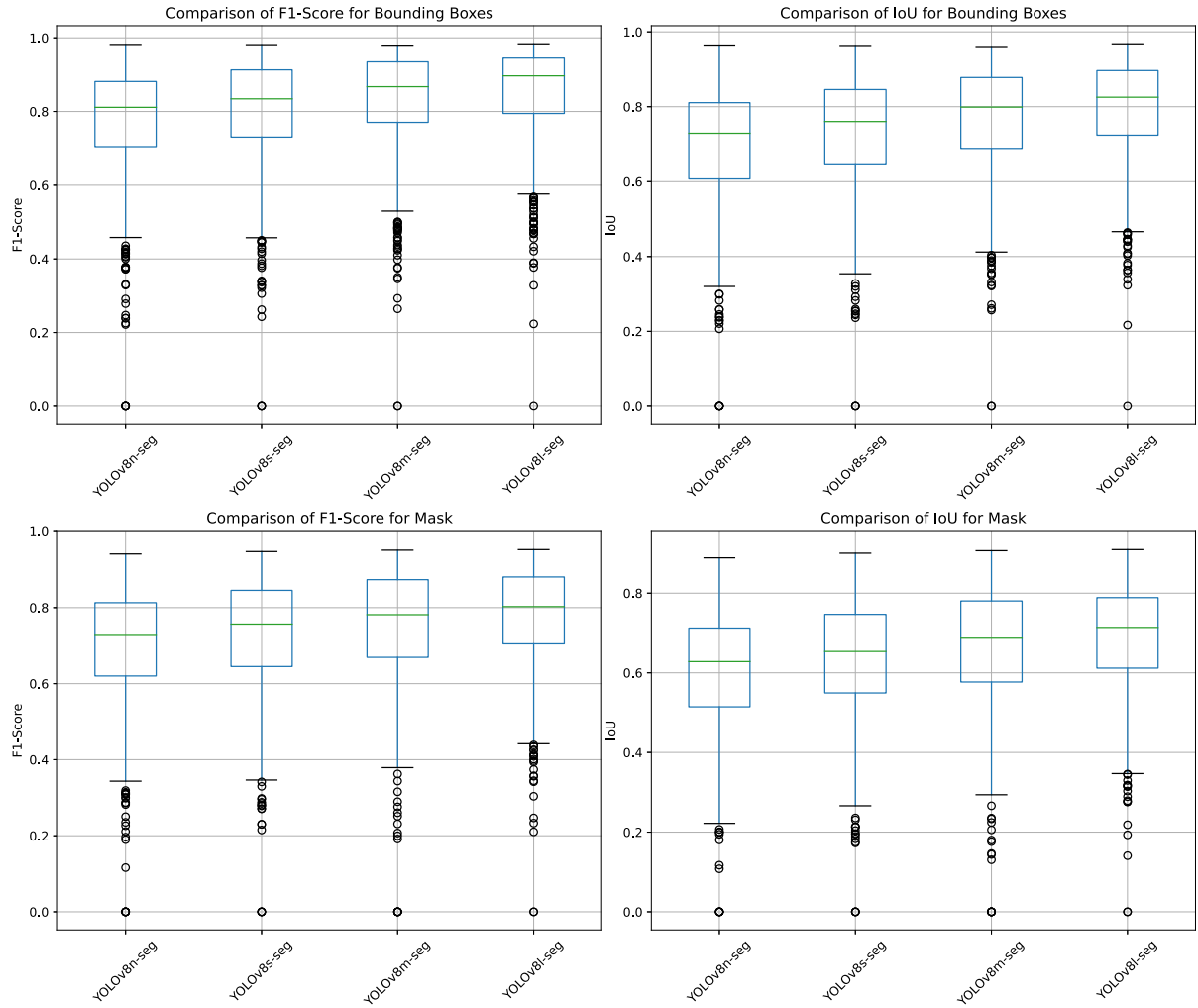


Fig. 9. Boxplots of F1-score and IoU metrics over the test dataset for YOLOv8-seg models.

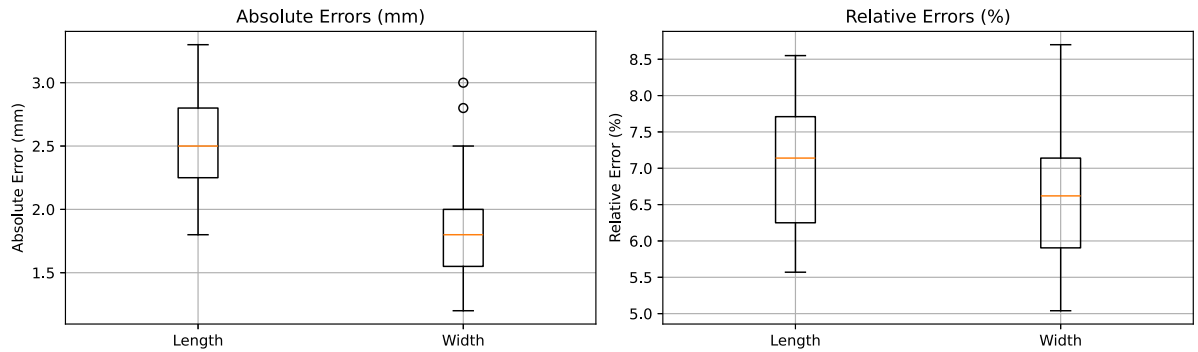


Fig. 10. Error measurements: Vernier caliper and computer vision algorithm.

real-time processing is prioritized, for example in robotic harvesting or fast in-field assessment, the lighter variants, particularly YOLOv8s-seg, offer a favorable compromise between accuracy and speed, with competitive F1-Score and IoU and more than 30 FPS for bounding boxes. Conversely, in offline analysis scenarios, where throughput is less critical and the main goal is to maximize the quality of downstream stages, namely feature extraction, rule-based size-maturity categorization, and yield mapping, the larger models, YOLOv8m-seg and YOLOv8l-seg provide clear gains in box and mask overlap, which translate into more precise fruit localization and contour delineation.

3.2. Evaluation of the accuracy of dimension measurements

The evaluation of the dimension estimation stage was carried out by comparing the length and width obtained by the vision algorithm with reference measurements taken with a Vernier caliper. To this end, 43 strawberries with representative sizes and shapes were selected, and for each fruit four values were recorded: length and width measured manually, and length and width estimated by the vision system. From these measurements, absolute errors (in mm) and relative errors (in %) were computed for both length and width, as summarized in Table 3.

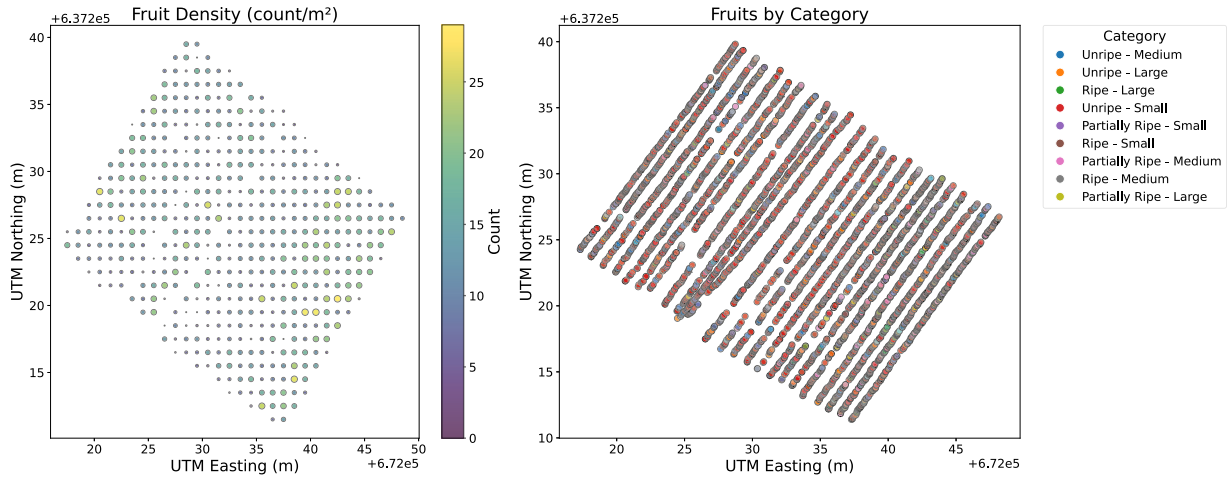


Fig. 11. Yield map: Fruit density and spatial distribution of operational size-maturity categories.

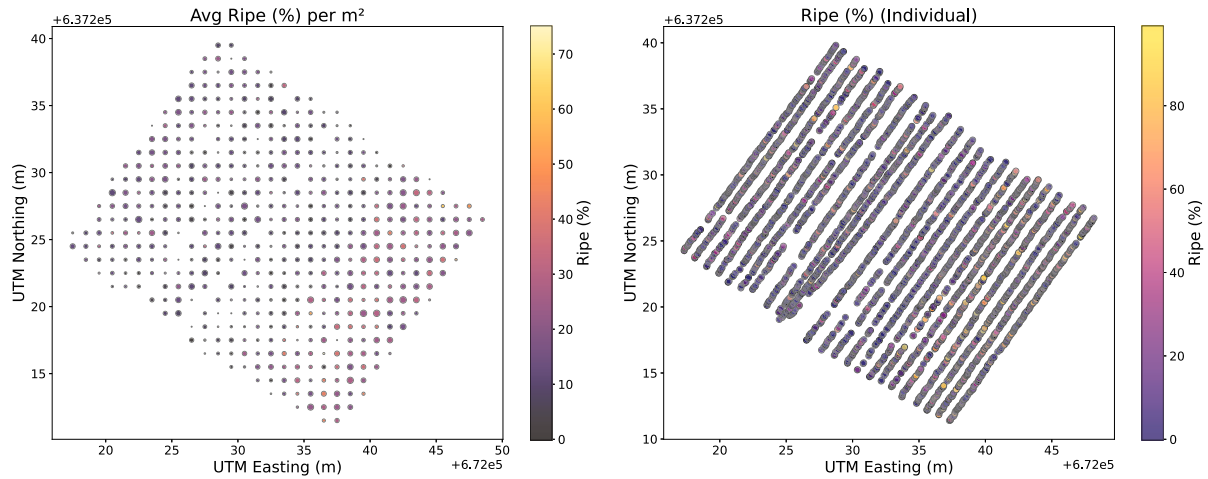


Fig. 12. Yield map: Average vs. individual ripeness.

The values reported for the computer vision algorithm are presented as integers because the calibrated estimates were rounded to the nearest millimeter, as defined in the dimension estimation stage. This decision avoids reporting sub-millimetric precision that is not justified under field conditions with an RGB-D sensor. In contrast, Vernier caliper measurements are kept with decimal resolution because they correspond to the manual reference measurements.

Table 3 shows that, for most fruits, the vision algorithm tends to slightly overestimate the dimensions with respect to the Vernier caliper. For length, absolute errors typically range between 2.0 and 3.0 mm, while for width they are usually between 1.2 and 2.5 mm. These values correspond to relative errors that, for length, are mostly between 5.5% and 8.5%, and for width are around 5%–8%. This systematic overestimation is consistent with the nature of the error sources in the vision system (depth quantization, sensor noise, point cloud discretization, and geometric projection) and indicates that the algorithm does not produce extreme random errors, but rather a moderate and relatively stable bias.

The data in the table are summarized graphically in Fig. 10, which presents two boxplot diagrams. The left panel shows the absolute errors in length and width (in mm), while the right panel shows the relative errors (in %). In both cases, the median absolute errors remain below 3 mm for length and below 2 mm for width, and the interquartile ranges are relatively compact, indicating a moderate dispersion of errors and

the absence of extreme outliers. Similarly, the boxplots of relative error show that most measurements are concentrated below 8% for both length and width, which is consistent with the values reported in the table.

From these individual measurements, global error metrics of the vision system were computed: Mean Absolute Error (MAE), Mean Relative Error (MRE), and Root Mean Squared Error (RMSE), for both length and width, using the definitions given in Section 2.3.2. The results, summarized in Table 4, indicate an MAE of 2.51 mm for length and 1.85 mm for width, with mean relative errors of 7.03% and 6.62%, respectively. The RMSE values are very close to the MAE (2.53 mm for length and 1.89 mm for width), which suggests that there are no isolated large errors dominating the global behavior; instead, the discrepancies between the caliper and the algorithm are distributed relatively uniformly across the sample.

To analyze the stability of the error with respect to fruit size, the dimensional range covered by the manual validation subset was examined. The evaluated strawberries had reference lengths between 30.4 and 42.0 mm and widths between 22.5 and 38.2 mm. Within this interval, the absolute errors remained within compact ranges: 1.8–3.3 mm for length and 1.2–3.0 mm for width. Similarly, the relative errors remained bounded between 5.57–8.55% for length and 5.04–8.70% for width. This indicates that, within the evaluated size range, the method maintains stable behavior and does not exhibit extreme errors associated with smaller or larger fruits.

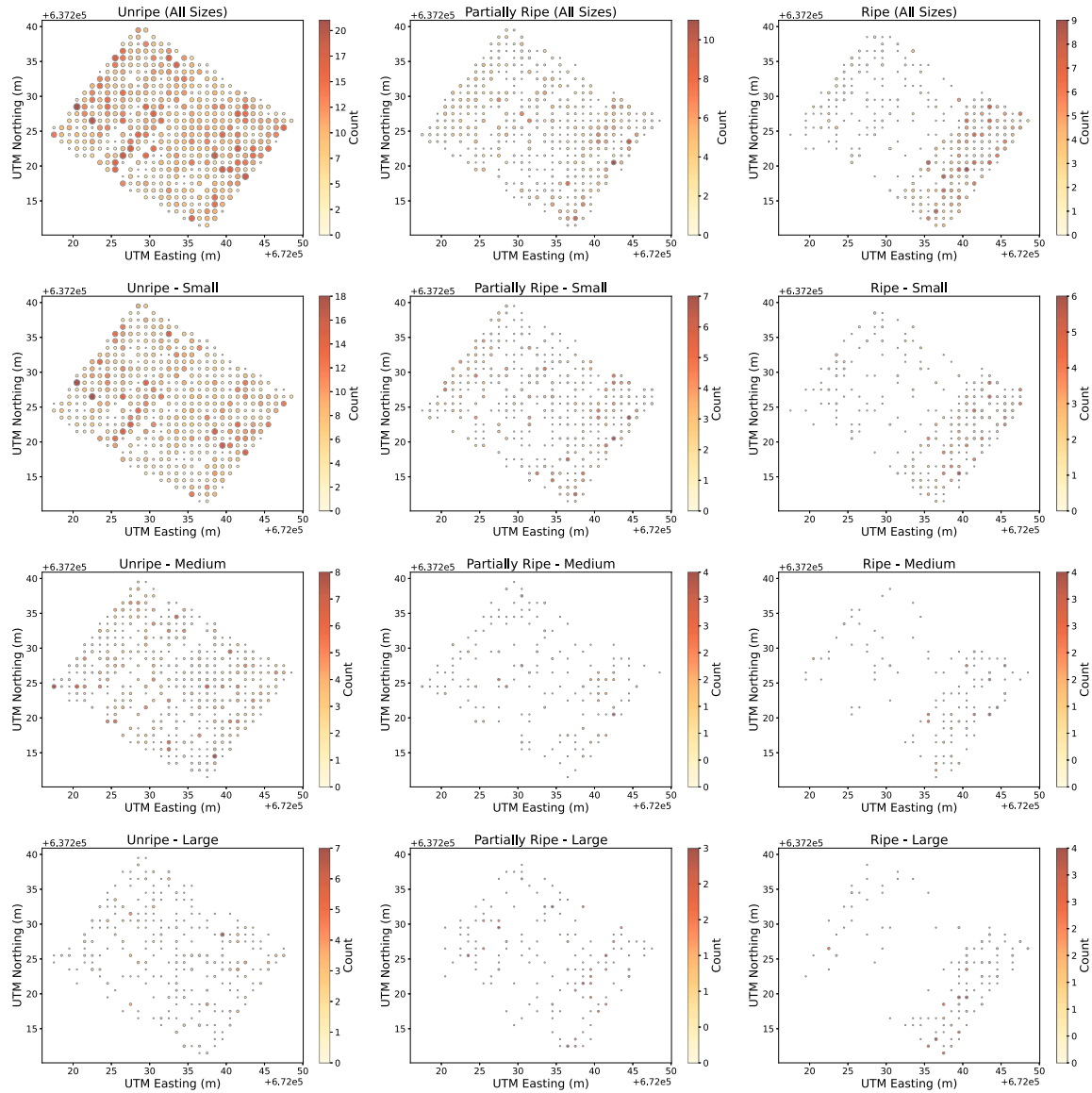


Fig. 13. Yield map: Fruit count by size and maturity class.

These results are coherent with the design of the dimension estimation stage, which combines depth maps, projection to metric coordinates, and point cloud filtering via DBSCAN. The removal of spurious points, especially those far from the main fruit body, prevents small noisy regions or background reflections from artificially inflating the maximum distances used to compute length and width. Even so, size estimation may be affected by depth quantization, sensor spatial resolution, small segmentation inaccuracies, partial occlusions, and incomplete fruit visibility from a lateral viewpoint. These factors explain the slight systematic overestimation observed and suggest that, in future validations with explicit occlusion annotations, the sensitivity of the method to partially visible fruits could be assessed in greater detail.

From an agronomic perspective, the observed errors are small relative to the typical dimensions of the strawberries analyzed, with lengths on the order of 30–40 mm and widths around 25–30 mm. The mean absolute error close to 2–3 mm indicates that the vision algorithm can reproduce the reference measurements with good accuracy under the evaluated conditions. Consequently, the accuracy achieved by the system is sufficient to characterize fruit size at the individual level and to support the subsequent stages of operational size–maturity categoriza-

tion and yield map generation, incorporating average size information and category composition across different zones of the crop.

3.3. Yield maps and spatial distribution of size–maturity categories

The yield maps generated from the complete set of georeferenced strawberries make it possible to move from point-based measurements to an explicitly spatial view of the crop. Unlike global analyses of total counts or category distributions, these maps show how fruit density, visible ripeness, and operational size–maturity categories are spatially organized across the field, revealing heterogeneities that are difficult to perceive by direct visual inspection.

3.3.1. Fruit density and category distribution

The first group of maps is based on the count of fruits per 1 m² cell, as defined in the yield mapping stage. Fig. 11 summarizes these results. In the left panel, the *Fruit Density* (count/m²) map shows, for each grid cell, the number of strawberries detected within that area, represented by symbols whose size and color are proportional to the local fruit count.

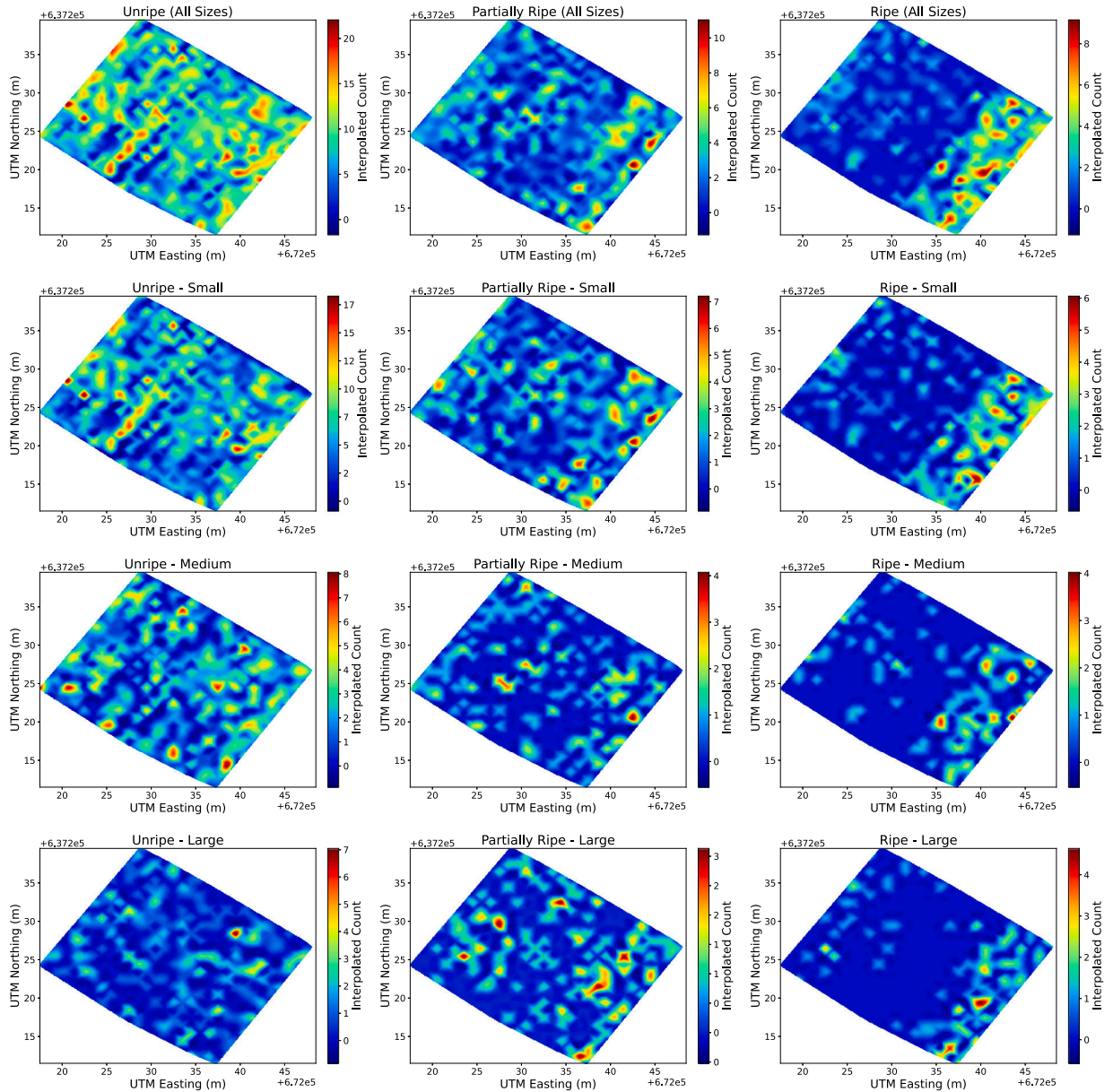


Fig. 14. Yield map: Interpolated fruit count by size and maturity class.

This map confirms a strong spatial variability: zones with high fruit concentration alternate with areas of lower density within the same plot.

The right panel of Fig. 11, labeled *Fruits by Category*, shows individual fruits at their UTM coordinates, colored according to the nine operational size–maturity categories defined in Section 2.4. This representation not only confirms the location of high-load zones, but also reveals which types of fruits dominate in each sector. In several dense regions, most strawberries belong to the “Unripe–Small” or “Unripe–Medium” categories, suggesting strong future production potential rather than immediate harvest availability. Conversely, there are patches with moderate density but a higher presence of “Ripe–Medium” and “Ripe–Large” fruits, which correspond to zones of particular interest for early or selective harvesting. The joint analysis of density and category maps allows areas with high fruit load but still immature fruits to be distinguished from less dense areas containing fruits closer to harvest.

3.3.2. Average vs. individual ripeness

Fig. 12 compares two complementary ways of representing fruit maturity. In the left panel, the *Average Ripeness per Cell* map associates each 1 m² grid cell with the mean ripe-area percentage of the fruits it contains, as defined in Eq. (26). This produces zones of higher or lower average ripeness, which can be interpreted as zones of the field that are phenologically ahead of or behind the rest of the plot.

The right panel shows the *Individual Ripeness* map, where each strawberry is displayed at its georeferenced position and colored according to its ripeness percentage, using the same color scale as in the cell-based map. It becomes clear that a cell with high average ripeness may be composed almost entirely of mature fruits, but it may also correspond to a mixture of very ripe strawberries with others that are still green. Likewise, some cells with intermediate average ripeness contain a mosaic of states, in which unripe, partially ripe, and ripe fruits coexist.

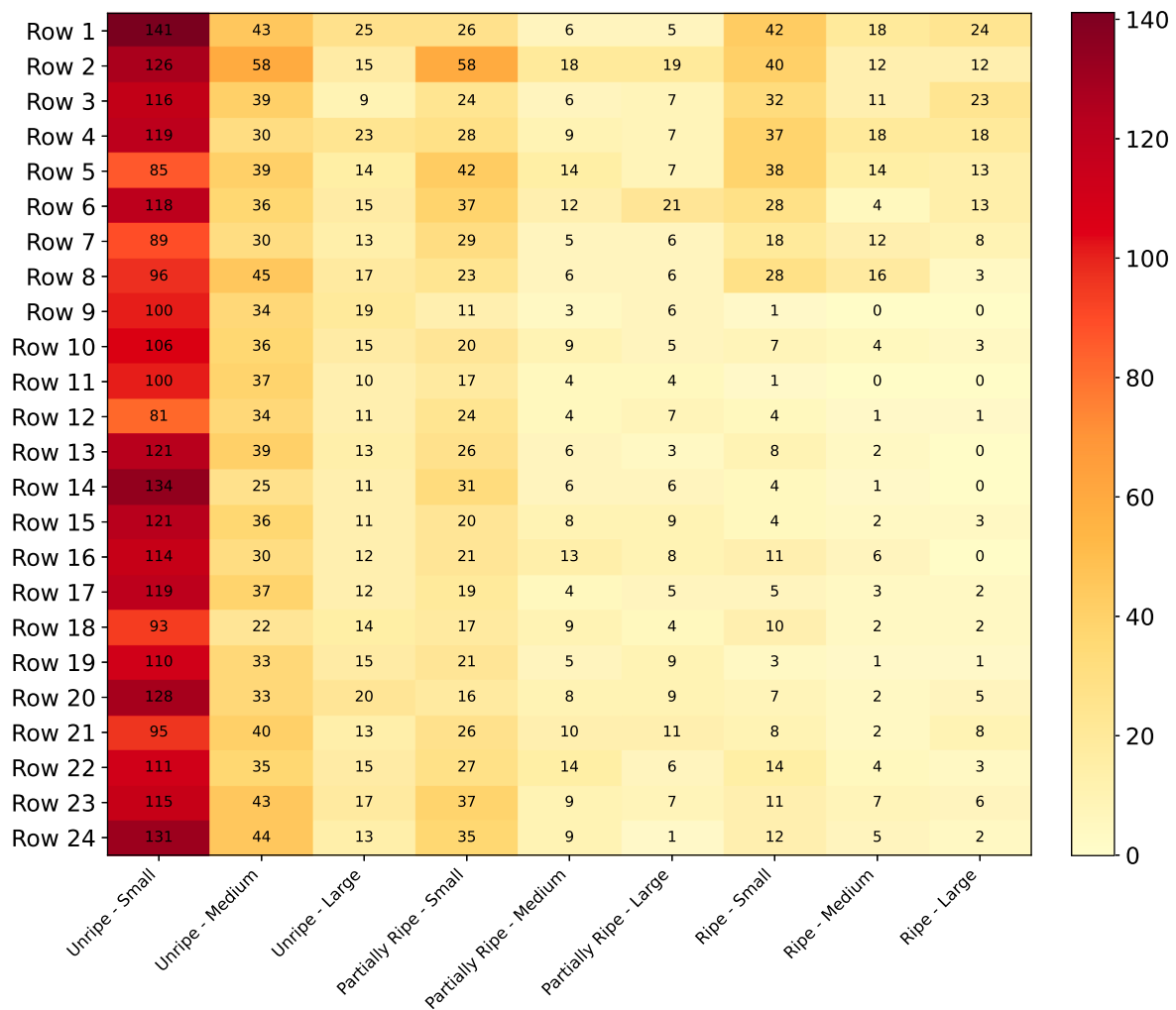


Fig. 15. Row-wise distribution of fruits by operational size-maturity category.

This comparison highlights the importance of considering both aggregated indicators (cell averages) and individual detail. The former is useful to quickly prioritize sectors of the plot, while the latter warns about internal heterogeneity and helps decide whether a partial harvest or a more gradual management strategy is advisable in specific areas.

3.3.3. Fruit count by size and maturity class

A third group of maps explores the interaction between size and maturity. Fig. 13 presents a 4×3 arrangement of yield maps. The three columns correspond to the aggregated maturity classes *Unripe*, *Partially Ripe*, and *Ripe*, while the four rows represent *All sizes*, *Small*, *Medium*, and *Large*, respectively.

In the first row (*All sizes*), the maps show the global spatial distribution of unripe, partially ripe, and ripe fruits per square meter. As expected from the category statistics, unripe fruits dominate large portions of the plot, consistent with the overall phenological state of the crop. However, when moving to the second, third, and fourth rows (*Small*, *Medium*, and *Large*), important differences emerge. In the “Unripe–Small” maps, extended bands with very high counts appear, indicating zones where fruit load is high but still at early stages of development. In contrast, the “Ripe–Large” maps reveal more localized patches, typically in sectors where plant vigor and microenvironmental conditions have favored more complete fruit development.

These maps support decisions such as: (i) prioritizing for harvest those areas where high density coexists with a predominance of ripe medium or large fruits; (ii) identifying zones that should be left for a

later harvest (areas dominated by “Unripe–Small” categories); and (iii) recognizing sectors that might require specific agronomic interventions to homogenize the crop status.

Complementarily, Fig. 14 shows the same 4×3 arrangement of size-maturity combinations, but using interpolated surfaces instead of discrete cells. Each map represents a continuous color field obtained by spatial interpolation of the counts per 1 m^2 cell. This continuous representation smooths transitions and makes spatial gradients in fruit concentration more evident. In particular, it helps identify zones of high density for specific size-maturity combinations and zones of gradual transition where the composition of the fruit changes along the plot. These interpolated maps are used as visual aids to interpret spatial trends; quantitative interpretation is based on the original counts computed in the 1 m^2 grid cells, since interpolation may produce smoothed or intermediate values that do not correspond to direct fruit-count observations.

3.3.4. Row-wise distribution of fruits by category

Finally, the row-wise organization of operational size-maturity categories is summarized in Fig. 15, which presents a heatmap-style visualization of *Fruits per Row by Category*. On the vertical axis, each row of the hydroponic system is represented, and on the horizontal axis the nine operational categories defined in Section 2.4 appear. The color intensity of each cell encodes the number of fruits of the corresponding category in that row.

Table 3
Comparison measurements: Vernier caliper and computer vision algorithm.

Fruit	Vernier Caliper		Computer Vision Algorithm		Absolute Error		Relative Error	
	Length (mm)	Width (mm)	Length (mm)	Width (mm)	Length (mm)	Width (mm)	Length (%)	Width (%)
1	37.0	28.0	40.0	30.0	3.0	2.0	8.11	7.14
2	31.0	25.5	33.0	27.0	2.0	1.5	6.45	5.88
3	36.0	25.3	39.0	27.0	3.0	1.7	8.33	6.72
4	35.8	22.5	38.0	24.0	2.2	1.5	6.15	6.67
5	32.0	23.5	34.0	25.0	2.0	1.5	6.25	6.38
6	34.1	23.8	36.0	25.0	1.9	1.2	5.57	5.04
7	33.0	27.2	35.0	29.0	2.0	1.8	6.06	6.62
8	34.8	26.1	37.0	28.0	2.2	1.9	6.32	7.28
9	33.5	28.2	36.0	30.0	2.5	1.8	7.46	6.38
10	32.6	25.7	35.0	27.0	2.4	1.3	7.36	5.06
11	37.6	27.1	40.0	29.0	2.4	1.9	6.38	7.01
12	34.2	29.0	37.0	31.0	2.8	2.0	8.19	6.90
13	37.7	36.0	40.0	39.0	2.3	3.0	6.10	8.33
14	34.5	29.5	37.0	32.0	2.5	2.5	7.25	8.47
15	33.0	27.8	35.0	30.0	2.0	2.2	6.06	7.91
16	33.3	27.3	36.0	29.0	2.7	1.7	8.11	6.23
17	32.0	23.0	34.0	25.0	2.0	2.0	6.25	8.70
18	36.3	27.5	39.0	29.0	2.7	1.5	7.44	5.45
19	34.5	28.2	37.0	30.0	2.5	1.8	7.25	6.38
20	36.1	29.7	39.0	32.0	2.9	2.3	8.03	7.74
21	32.6	25.2	35.0	27.0	2.4	1.8	7.36	7.14
22	30.4	25.3	33.0	27.0	2.6	1.7	8.55	6.72
23	32.2	27.2	34.0	29.0	1.8	1.8	5.59	6.62
24	32.6	23.6	35.0	25.0	2.4	1.4	7.36	5.93
25	32.5	26.5	35.0	28.0	2.5	1.5	7.69	5.66
26	37.5	27.4	40.0	29.0	2.5	1.6	6.67	5.84
27	38.3	30.2	41.0	32.0	2.7	1.8	7.05	5.96
28	36.6	30.1	39.0	32.0	2.4	1.9	6.56	6.31
29	39.6	27.5	42.0	29.0	2.4	1.5	6.06	5.45
30	37.6	28.4	40.0	30.0	2.4	1.6	6.38	5.63
31	40.1	28.0	43.0	30.0	2.9	2.0	7.23	7.14
32	37.8	26.5	40.0	28.0	2.2	1.5	5.82	5.66
33	38.6	28.5	41.0	30.0	2.4	1.5	6.22	5.26
34	39.2	29.4	42.0	31.0	2.8	1.6	7.14	5.44
35	37.0	35.5	40.0	38.0	3.0	2.5	8.11	7.04
36	37.1	30.5	40.0	33.0	2.9	2.5	7.82	8.20
37	39.1	38.2	42.0	41.0	2.9	2.8	7.42	7.33
38	36.2	29.0	39.0	31.0	2.8	2.0	7.73	6.90
39	40.7	26.4	44.0	28.0	3.3	1.6	8.11	6.06
40	42.0	26.3	45.0	28.0	3.0	1.7	7.14	6.46
41	35.8	28.7	38.0	31.0	2.2	2.3	6.15	8.01
42	37.0	30.2	40.0	32.0	3.0	1.8	8.11	5.96
43	35.6	26.0	38.0	28.0	2.4	2.0	6.74	7.69

Table 4
Global error metrics for measurement comparison.

Metric	Value	Unit
Mean Absolute Error (Length)	2.51	mm
Mean Absolute Error (Width)	1.85	mm
Mean Relative Error (Length)	7.03	%
Mean Relative Error (Width)	6.62	%
RMSE (Length)	2.53	mm
RMSE (Width)	1.89	mm

This representation complements the 2D spatial maps with an operational perspective: it allows immediate identification of which rows concentrate the largest numbers of “Ripe–Medium” or “Ripe–Large” fruits as priority candidates for harvest, which rows are dominated by “Unripe–Small” fruits as zones still in development, and where more balanced mixtures of states and sizes occur. Moreover, comparing rows reveals possible management effects, for example differences in irrigation or fertigation between lines, and their impact on the distribution of size–maturity categories.

Overall, the different groups of yield maps, including global and category-specific density, average versus individual ripeness, size and maturity counts, and row-wise distributions, provide a rich geospatial characterization of the crop. Beyond quantifying how many strawber-

ries are present, they answer key questions about *where* harvest-priority fruits are located, *in what state* they are distributed across the plot, and *how* internal variability is structured, thus offering a robust basis for harvest planning and decision making in precision agriculture.

4. Conclusion

This work presented an integrated RGB-D and GNSS RTK pipeline for hydroponic strawberries that links instance segmentation, feature extraction, rule-based operational size–maturity categorization, and yield mapping at the individual fruit level. The YOLOv8-seg models provided fruit masks, bounding boxes, and detections with sufficient quality to feed the subsequent stages of size estimation, ripeness estimation, and spatial distribution analysis under real field conditions.

By projecting depth information into 3D and filtering point clouds with DBSCAN, the proposed method estimated strawberry length and width with mean absolute errors of 2.51 mm and 1.85 mm, respectively, relative to Vernier caliper measurements. These results indicate that the system can characterize fruit dimensions at the individual level with sufficient accuracy to support operational categorization and spatial aggregation. By combining these dimensions with visible ripeness percentage, each fruit was assigned to one of nine size–maturity categories using the predefined rule-based procedure described in the methodology.

The yield maps derived from georeferenced fruits revealed marked spatial variability in density, visible ripeness, and size-maturity composition, both at the 1 m² grid scale and along crop rows. These maps make it possible to identify zones with high concentrations of harvest-priority fruits, areas dominated by developing immature fruits, and spatial patterns potentially associated with management differences within the crop.

The system was implemented and applied as a processing pipeline on field-collected data. Although the evaluation was conducted in a single commercial hydroponic crop, the results show the potential of the approach as a decision-support tool for harvest planning in precision horticulture. Transfer to other cultivars, planting configurations, or environmental conditions will require recalibration of maturity and size thresholds, as well as renewed validation of the perception module. Future work should also compare more recent segmentation architectures and evaluate the system under different levels of occlusion, illumination, and crop density.

CRedit authorship contribution statement

Camilo Pardo-Beainy: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Carlos Parra:** Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Conceptualization; **Leonardo Solaque:** Writing – original draft, Supervision, Resources, Methodology, Conceptualization.

Ethical statement

This study did not involve human participants or animals, and therefore ethical approval was not required. All data were collected through field experiments conducted in accordance with relevant regulations. Artificial intelligence tools were used solely for language editing and did not influence the scientific content or conclusions of this manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] S. Sharmin, M.T. Hossan, M.S. Uddin, A review of machine learning approaches for predicting lettuce yield in hydroponic systems, 2025. <https://doi.org/10.1016/j.atech.2025.100925>

- [2] K.C. Swain, Q.U. Zaman, A.W. Schumann, D.C. Percival, D.D. Bochtis, Computer vision system for wild blueberry fruit yield mapping, *Biosyst. Eng.* 106 (2010) 389–394. <https://doi.org/10.1016/j.biosystemseng.2010.05.001>
- [3] A.L.S. Safre, A. Torres-Rua, B.L. Black, S. Young, Deep learning framework for fruit counting and yield mapping in tart cherry using YOLOv8 and YOLO11, *Smart Agric. Technol.* 11 (2025) 100948. <https://doi.org/10.1016/j.atech.2025.100948>
- [4] T. Tamilarasi, P. Muthulakshmi, Machine vision algorithm for detection and maturity prediction of brinjal, *Smart Agric. Technol.* 7 (2024). <https://doi.org/10.1016/j.atech.2024.100402>
- [5] F.X. Dono, B.N. Baatuwue, F.K. Abagale, P.B. Sørensen, Application of computer vision and machine learning in morphological characterization of Adansonia digitata fruits, *Smart Agric. Technol.* 9 (2024). <https://doi.org/10.1016/j.atech.2024.100528>
- [6] P. Chu, Z. Li, K. Zhang, K. Lammers, R. Lu, High-precision fruit localization using active laser-camera scanning: robust laser line extraction for 2D-3D transformation, *Smart Agric. Technol.* 7 (2024). <https://doi.org/10.1016/j.atech.2023.100391>
- [7] O.E. Apolo-Apolo, J. Martínez-Guante, G. Egea, P. Raja, M. Pérez-Ruiz, Deep learning techniques for estimation of the yield and size of citrus fruits using a UAV, *Eur. J. Agron.* 115 (2020) 126030. <https://doi.org/10.1016/j.eja.2020.126030>
- [8] A. Aquino, J.M. Ponce, M. Noguera, J.M. Andújar, Olive-fruit yield estimation by modelling perceptual visual features, *Comput. Electron. Agric.* 214 (2023) 108361. <https://doi.org/10.1016/j.compag.2023.108361>
- [9] L. Wang, K. Yang, S. Zeng, Y. Yi, B. Li, A novel stone fruit quality non-destructive evaluation method based on PVR by LF-NMRI, *J. Food Eng.* 387 (2025). <https://doi.org/10.1016/j.jfoodeng.2024.112338>
- [10] J. Giménez-Gallego, J. Martínez-del Rincon, J.D. González-Teruel, H. Navarro-Hellín, P.J. Navarro, R. Torres-Sánchez, On-tree fruit image segmentation comparing mask R-CNN and vision transformer models. application in a novel algorithm for pixel-based fruit size estimation, *Comput. Electron. Agric.* 222 (2024). <https://doi.org/10.1016/j.compag.2024.109077>
- [11] P. Roy, A. Kislal, P.A. Plonski, J. Luby, V. Isler, Vision-based preharvest yield mapping for apple orchards, *Comput. Electron. Agric.* 164 (2019). <https://doi.org/10.1016/j.compag.2019.104897>
- [12] Z. Li, R. Xu, C. Li, P. Munoz, F. Takeda, B. Leme, In-field blueberry fruit phenotyping with a MARS-PhenoBot and customized BerryNet, *Comput. Electron. Agric.* 232 (2025). <https://doi.org/10.1016/j.compag.2025.110057>
- [13] M. Churuvija, R. Sapkota, D. Ahmed, M. Karkee, A pose-versatile imaging system for comprehensive 3D modeling of planar-canopy fruit trees for automated orchard operations, *Comput. Electron. Agric.* 230 (2025). <https://doi.org/10.1016/j.compag.2025.109899>
- [14] A. Subeesh, S.P. Kumar, S.K. Chakraborty, K. Uppendar, N.S. Chandel, D. Jat, K. Dubey, R.U. Modi, M.M. Khan, UAV imagery coupled deep learning approach for the development of an adaptive in-house web-based application for yield estimation in citrus orchard, *Meas. J. Int. Meas. Confed.* 234 (2024). <https://doi.org/10.1016/j.measurement.2024.114786>
- [15] X. Zhang, A. Toudeshki, R. Ehsani, H. Li, W. Zhang, R. Ma, Yield estimation of citrus fruit using rapid image processing in natural background, *Smart Agric. Technol.* 2 (2022). <https://doi.org/10.1016/j.atech.2021.100027>
- [16] L. He, W. Fang, G. Zhao, Z. Wu, L. Fu, R. Li, Y. Majeed, J. Dhupia, Fruit yield prediction and estimation in orchards: a state-of-the-art comprehensive review for both direct and indirect methods, 2022. <https://doi.org/10.1016/j.compag.2022.106812>
- [17] C. Pardo-Beainy, C. Parra, L. Solaque, W.S. Lee, Deep learning and georeferenced RGB-D imaging for hydroponic strawberry yield mapping, *Smart Agric. Technol.* 12 (2025) 101293. <https://doi.org/10.1016/j.atech.2025.101293>
- [18] T. Thayananthan, X. Zhang, Y. Huang, J. Chen, N.K. Wijewardane, V.S. Martins, G.D. Chessier, C.T. Goodin, CottonSim: a vision-guided autonomous robotic system for cotton harvesting in Gazebo simulation, *Comput. Electron. Agric.* 239 (2025) 110963. <https://doi.org/10.1016/j.compag.2025.110963>
- [19] Y. Zheng, A. Liu, A. Huang, D.-H. Kim, Y. Shen, K.-H. Lee, Realtime multi-RGBD SLAM framework for 3D reconstruction and phenotyping in large-scale apple orchards, *Comput. Electron. Agric.* 240 (2026) 111170. <https://doi.org/10.1016/j.compag.2025.111170>
- [20] M.N. Conejero, H. Montes, J.M. Bengochea-Guevara, L. Garrido-Rey, D. Andújar, A. Ribeiro, A collaborative robotic fleet for yield mapping and manual fruit harvesting assistance, *Comput. Electron. Agric.* 235 (2025) 110351. <https://doi.org/10.1016/j.compag.2025.110351>
- [21] A.F. Colaço, R.G. Trevisan, F.H.S. Karp, J.P. Molin, Yield mapping methods for manually harvested crops, *Comput. Electron. Agric.* 177 (2020). <https://doi.org/10.1016/j.compag.2020.105693>
- [22] M.K. Verma, M. Yadav, 3D LiDAR-based techniques and cost-effective measures for precision agriculture: a review, *Rev. Int. Geomat.* 34 (1) (2025) 855–879. <https://doi.org/10.32604/riig.2025.069914>
- [23] K.M.S. Haque, P. Joshi, N. Subedi, Integrating UAV-based multispectral imaging with ground-truth soil nitrogen content for precision agriculture: a case study on paddy field yield estimation using machine learning and plant height monitoring, *Smart Agric. Technol.* 12 (2025) 101542. <https://doi.org/10.1016/j.atech.2025.101542>